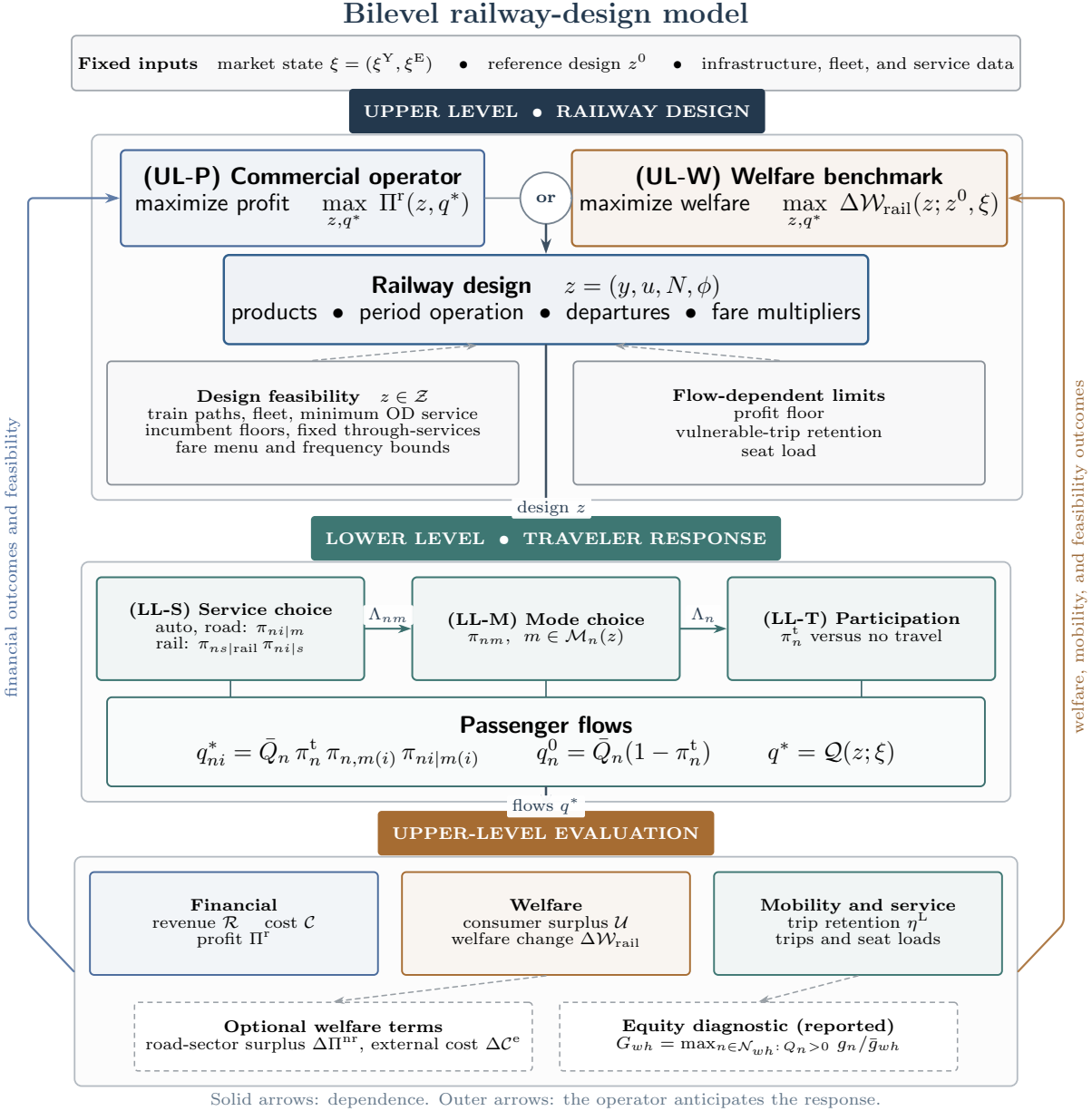

Bilevel railway-design model
Fixed inputs market state $\xi = (\xi^{Y}, \xi^{E})$ • reference design z^0 • infrastructure, fleet, and service data
UPPER LEVEL • RAILWAY DESIGN
(UL-P) Commercial operator
maximize profit $\max_{z,q^*} \Pi^{r}(z, q^*)$
or
(UL-W) Welfare benchmark
maximize welfare $\max_{z,q^*} \Delta\mathcal{W}_{rail}(z; z^0, \xi)$
Railway design $z = (y, u, N, \phi)$
products • period operation • departures • fare multipliers
Design feasibility $z \in \mathcal{Z}$
train paths, fleet, minimum OD service
incumbent floors, fixed through-services
fare menu and frequency bounds
Flow-dependent limits
profit floor
vulnerable-trip retention
seat load
design z
LOWER LEVEL • TRAVELER RESPONSE
(LL-S) Service choice
auto, road: $\pi_{ni|m}$
rail: $\pi_{ns|rail}\,\pi_{ni|s}$
Λ_{nm}
(LL-M) Mode choice
π_{nm}, $m \in \mathcal{M}_n(z)$
Λ_n
(LL-T) Participation
π^{t}_{n} versus no travel
Passenger flows
$q^*_{ni} = \bar{Q}_n\,\pi^{t}_{n}\,\pi_{n,m(i)}\,\pi_{ni|m(i)}$ $q^0_n = \bar{Q}_n(1 - \pi^{t}_{n})$ $q^* = \mathcal{Q}(z; \xi)$
flows q^*
UPPER-LEVEL EVALUATION
Financial
revenue $\mathcal{R}$ cost $\mathcal{C}$
profit Π^{r}
Welfare
consumer surplus $\mathcal{U}$
welfare change $\Delta\mathcal{W}_{rail}$
Mobility and service
trip retention η^{L}
trips and seat loads
Optional welfare terms
road-sector surplus $\Delta\Pi^{nr}$, external cost $\Delta\mathcal{C}^{e}$
Equity diagnostic (reported)
$G_{wh} = \max_{n \in \mathcal{N}_{wh}:\, Q_n > 0} g_n/\bar{g}_{wh}$
financial outcomes and feasibility
welfare, mobility, and feasibility outcomes
Solid arrows: dependence. Outer arrows: the operator anticipates the response.